

Full-production discretization P3: independent continuum quadrature

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

This supplement closes the executable independent-reference part of P3 item 1. It does not identify a same-backend fine-grid residual with the continuum residual. Instead it prolongs a declared analytic field from S_N to two oversampling grids, differentiates an independently implemented portable trapezoid-quadrature functional, and projects the result back to S_N .

1 Independent quadrature proxy

Let $I_M : S_N \rightarrow S_M$ be Fourier zero-padding and P_N the real- L^2 Fourier projector. The computed reference is

$$R_{N,M}^{\text{IQ}}(\Psi_N) = P_N R_M^P(I_M \Psi_N),$$

where R_M^P is the real gradient of the portable functional implementation. For the analytic, nonzero-density manufactured field, the density floor keeps all rational Class-II coefficients smooth. Periodic trapezoid quadrature then converges superalgebraically for smooth periodic integrands and exponentially for this analytic test input. The M24/M32 difference is retained as observed reference uncertainty, not hidden as exact continuum data.

2 Executed result

For N8, N12, and N16, exact Fourier prolongation to M24/M32 differs from direct analytic sampling by at most $3.0848 \cdot 10^{-16}$. The portable and canonical gradients agree exactly on every tested target and oversampling grid. The projected M24/M32 uncertainty is at most $2.5180 \cdot 10^{-14}$. The canonical collocation residual differs from the independent projected proxy by

$$5.3301196 \cdot 10^{-6}, \quad 4.2482266 \cdot 10^{-10}, \quad 6.6668345 \cdot 10^{-12},$$

with observed orders 23.2750 and 14.4414. All 11 assertions pass.

3 Devil's-advocate and mandatory adversarial review

Sign and factor. The portable code differentiates its own real functional in explicit real coordinates; it does not reuse the canonical residual. The canonical comparison then detects a factor or pairing mismatch. **Convention.** Fourier prolongation divides forward coefficients by the source count and multiplies inverse reconstruction by the target count; the same centered convention is used by P_N . **Units.** The fixed torus and dimensionless P1 normalization are unchanged. **Convergence.** Two independent oversampling resolutions measure the quadrature tail. **Hardcode masking.** No measured residual, rate, or reference value is an input; grids and thresholds are manifest inputs. **Limit cases.** This audit is analytic, nonzero-density, eta-shell-zero, and CPU complex128 only. It does not bound arbitrary H2 P2 data at time

zero, floor removal, GPU behavior, or a historical solver. External review of the periodic quadrature-to-continuum passage is explicitly invited.

4 Remaining analytic passage

P2 gives C^∞ regularity for every positive time, but the present result does not yet supply a uniform Sobolev/analytic-norm bound on a P2 solution ball that turns the observed quadrature tail into a theorem. That bound, plus a dealiased/oversampled evolution comparison and the CUDA matrix, is required before Sector-B solver outputs may be described as controlled continuum PDE approximations.

5 Reproduction

Run `python codes/foundations/a3_full_production_independent_galerkin.py`. Expected: 11/11 PASS and A3-FULL-INDEPENDENT-GALERKIN-PASS.

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Independent analytic manufactured-field quadrature proxy converges and agrees with canonical collocation.
Scope:	Fixed T3, pinned functional and floors, eta_shell=0, analytic nonzero-density field, N8/12/16, M24/32.
Dependencies:	P1 canonical backend; P2 well-posed PDE package.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL; T3 scaffold.
Reproduction command:	<code>python codes/foundations/a3_full_production_independent_galerkin.py</code>
Expected output:	11/11 PASS; A3-FULL-INDEPENDENT-GALERKIN-PASS.
Falsification gate:	A3-FULL-DISCRETIZATION-CLOSURE.
Tier before / after:	T3 / T3.
No-overclaim statement:	Not a uniform P2 solution-ball bound, GPU execution, historical-solver result, T5, T6, or T7.
Next required action:	Uniform analytic Galerkin error bound, CUDA rows, and integrated one-command verification.